From Biomedical Text to Structured Knowledge in the Gut-Brain Axis

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Abstract

The exponential growth of scientific publications has increased the problem of data overload, particularly in the biomedical domain, challenging researchers and clinicians in staying updated and interpreting findings in large volumes of complex and specialized literature. In such settings, Natural Language Processing (NLP) methods for Information Extraction (IE) are essential to automatically convert unstructured texts into structured, machine-actionable knowledge. This need is especially pronounced in fast-evolving research areas such as the gut-brain axis, which explores the interplay between gut microbiota and neurological or psychiatric disorders. Despite significant progress in biomedical NLP, existing IE benchmarks often suffer from limited scope, coarse-grained annotation schemes, or heavy reliance on distantly supervised data. These factors severely constrain the robustness and effectiveness of trained IE systems. To address these limitations, we introduce GutBrainIE, a comprehensive IE benchmark built on more than 1,600 PubMed abstracts manually annotated with named entities, concept-level links, and relations under the most comprehensive and fine-grained annotation schema to date in IE. These features allow the benchmark to support 3 core NLP tasks: Named Entity Recognition (NER), Named Entity Linking (NEL), and Relation Extraction (RE). While centered on the gut-brain axis, GutBrainIE is designed as a general, reusable resource that provides a challenging and reliable framework for the development and evaluation of biomedical IE systems across domains. Building on these annotations, we also introduce GutBrainKB, a Knowledge Base (KB) that integrates the extracted information into a semantically structured knowledge graph aligned with a domain ontology and enriched with textual and bibliographic evidence, facilitating the exploration and exploitation of the knowledge embedded within GutBrainIE.

Keywords

Information Extraction (IE), Natural Language Processing (NLP), Named Entity Recognition (NER), Named Entity Linking (NEL), Relation Extraction (RE), Knowledge Graphs (KGs), Knowledge Bases (KBs), Gut-Brain Axis

1. Introduction

The volume and complexity of biomedical scientific literature are growing at an unprecedented pace, particularly in emerging research fields such as the gut-brain axis, where new findings linking gut microbiota to neurological and psychiatric disorders are continuously published in scientific articles, as shown in Figure 1 [1, 2, 3, 4]. This rapid growth poses a challenge for researchers and clinicians in identifying and interpreting relevant findings across large collections of unstructured scientific texts.

Natural Language Processing (NLP) techniques for Information Extraction (IE) provide a way to address this challenge by automatically transforming unstructured textual data into structured representations. Core IE tasks include Named Entity Recognition (NER), Named Entity Linking (NEL), and

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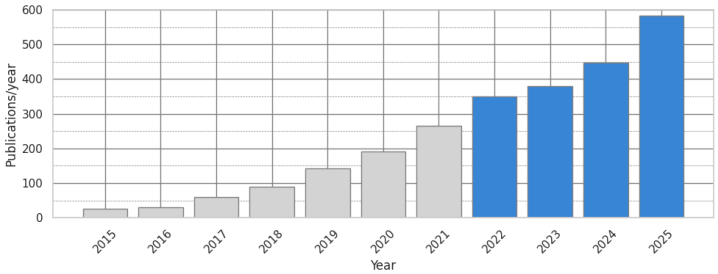


Figure 1: Annual rate of biomedical publications on PubMed related to the gut-brain axis, retrieved with the query *"gut microbiota[Title]" AND ("Parkinson[Title]" OR "mental health[Title]")* on February 19, 2026. Blue bars highlight years with more than 300 records.

Relation Extraction (RE). NER involves detecting and classifying mentions of biomedical entities such as diseases, drugs, or bacteria. NEL disambiguates and normalizes these entities by associating them to unique identifiers in structured Knowledge Bases (KBs). RE identifies and classifies semantic relations among entities, such as interactions or cause-effect associations [5].

The effectiveness of IE systems strongly depends on the availability of high-quality annotated datasets for training and evaluation. However, in the biomedical domain, creating such resources is expensive and time-consuming due to the need for domain expertise and the complexity of biomedical terminology. Consequently, existing corpora are typically restricted in scope to small, limited sets of entity types or relation predicates, contain relations annotated only at the sentence level, and often depend on noisy, distantly supervised annotations. [6, 7, 8]. These limitations hinder the development of IE systems capable of handling domains with highly specialized terminology and concept diversity, necessitating deep contextual and semantic understanding [9, 10]. Moreover, scientific texts often contain relations spanning multiple sentences, making sentence-level annotations insufficient for training robust IE systems [11, 12]. The gut-brain axis literature exemplifies these challenges, with its specialized terminology, diverse concepts, and frequent cross-sentence relations [13, 14].

To address these gaps, we developed GutBrainIE, a curated benchmark for biomedical IE focused on the gut-brain axis. The dataset includes fine-grained annotations for entity mentions, concept-level links, and semantic relations extracted from PubMed abstracts, thus supporting all three core IE tasks mentioned above: NER, NEL, and RE [15, 16]. These structured annotations also enable the construction of KBs, supporting a complete and in-depth exploration of the information and knowledge embedded within the corpus. Accordingly, we constructed GutBrainKB, a KB derived from the annotations of the GutBrainIE collection. GutBrainKB represents entities and relations as RDF triples aligned with a domain ontology derived from the GutBrainIE annotation schema, enabling structured exploration of gut-brain interactions and facilitating integration with other biomedical knowledge resources [17].

In this paper, we present the GutBrainIE benchmark and the GutBrainKB knowledge base. We describe the dataset's design and curation, analyze its features, and compare it with established, widely used IE datasets from both the biomedical and general-purpose domains. We then introduce the associated benchmark tasks and discuss their validation. Finally, we present the pipeline used to construct the GutBrainKB, the resulting structured knowledge graph, and the web application we implemented to facilitate its navigation and exploration.

2. The GutBrainIE Benchmark

The core part of the GutBrainIE benchmark is its annotated collection, a biomedical corpus containing both manually curated and automatically generated annotations for entity mentions, concept-level links, and relations extracted from PubMed abstracts related to the gut-brain axis.

The annotation schema includes the 13 entity types shown in Table 1, covering general biomedical concepts, entities specific to the gut-brain axis domain, and labels specific to medical experiments. For RE, the schema defines 17 relation predicates. Several predicates are overloaded, meaning that they

can connect different combinations of entity types depending on the context. This design results in 55 distinct relation triples (cf. Table 6 in the annex). In addition to entity and relation annotations, the collection features NEL annotations, where mentions are mapped to concepts drawn from six biomedical vocabularies, while unmatched mentions are associated with a custom ontology. The reference vocabularies considered for each entity type can be found in Table 5.

Given the specialized nature of the gut-brain axis domain, the annotation process relied on an in-house team rather than external crowdworkers. The team consisted of 7 expert annotators, including computer scientists, medical-domain terminologists, and biomedical specialists, along with 40 trained master's students in terminography acting as lay annotators. This choice enabled targeted training, continuous feedback, and greater consistency compared to fully crowdsourced annotation.

The dataset was constructed through the four-stage workflow depicted in Figure 2. Documents were first retrieved from PubMed using queries targeting studies on gut microbiota and its implications in neurological and mental disorders. After filtering duplicates and low-relevance publications, the final corpus consisted of 1,647 abstracts. The annotation schema and guidelines were then developed by manually analyzing a representative subset of documents with the support of biomedical specialists and terminologists, resulting in the final definition of entity types and relation predicates.

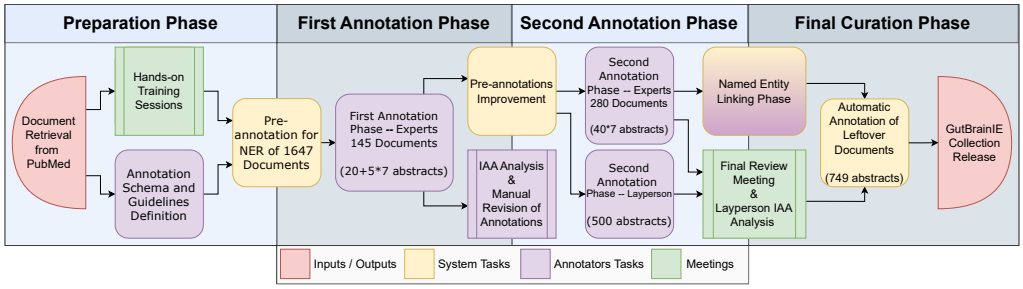


Figure 2: The four-stage workflow employed for creating the GutBrainIE corpus.

Manual annotation was carried out in two phases. In the first phase, expert annotators curated entity mentions and relations with the support of automatic NER pre-annotations, generated using the GLiNER model [18]. To assess the practical benefits of this pre-annotation approach, two terminologists were asked to annotate twenty documents divided into two batches, one with pre-annotations and the other without. Measurements showed that, without pre-annotations, they needed approximately 30 minutes per document to fully annotate its entity mentions. With pre-annotations, the time dropped to 10-15 minutes, depending on the document's complexity. To monitor annotation consistency, all expert annotators were assigned a subset of shared documents, enabling the computation of Inter-Annotator Agreement (IAA) [19, 20]. The resulting agreement scores were high for NER (Fleiss' $\kappa = 0.89$ and mean pairwise Cohen's $\kappa = 0.88$), comparable with other biomedical datasets [21, 22]. Agreement for RE was lower (0.43 for both metrics) but still consistent with prior works involving complex relation annotation, reflecting the semantic difficulty of the gut-brain axis domain [23]. A subsequent expert revision step validated the entire set of annotations produced in this phase.

In the second annotation stage, expert annotators continued the curation while trained lay annotators, working under expert supervision, were introduced to increase annotation throughput. Among the documents assigned to lay annotators, we included a subset annotated by experts in the previous annotation phase. These shared documents were used to assess annotation quality by computing IAA (Cohen's κ) against the expert-curated reference annotations [24]. IAA was moderate for NER ($\kappa = 0.50$) and lower for RE ($\kappa = 0.17$), confirming that non-expert annotators can reliably support entity annotation while relation curation demands more domain-specific expertise.

After completing the manual annotation stages, NEL was performed on the expert-curated mentions, as laypeople annotations often lack the precision needed for reliable concept mapping. The linking procedure employed is a three-stage sequential pipeline combines, in order, normalization heuristics, exact matching against reference vocabularies, and semantic similarity using contextual embeddings

Table 1
Overview of the 13 entity labels used in the GutBrainIE corpus, grouped according to their semantic role in the annotation schema. *DDF* stands for "Disease, Disorder, or Finding".

Category	Entity Labels
General biomedical	Anatomical Location, Animal, Bacteria, Chemical, DDF, Drug, Gene, Human
Gut-brain axis specific	Dietary Supplement, Food, Microbiome
Experimental specific	Biomedical Technique, Statistical Technique

from BiomedBERT [25]. When no candidate concept is found in the reference vocabularies, mentions are linked to a custom ontology, which was extended with new concepts when necessary. The quality of NEL annotations was validated by experts using a sampling-based evaluation framework with statistical guarantees, which gave an estimated overall accuracy of 0.915 ± 0.0473 [26].

To expand the dataset, the remaining unannotated documents were labeled automatically. For NER, GLiNER was fine-tuned on the expert-curated annotations, while RE was performed using the ATLOP model, which is particularly effective in capturing long-tail relations, a typical scenario in biomedical literature [18, 27]. Although these annotations were not manually revised, post-processing heuristics were applied to ensure consistency with the annotation schema and guidelines.

Finally, the corpus was organized into four quality-based folds reflecting the reliability of the annotations: Platinum (expert annotations reviewed by a dedicated subgroup of experts), Gold (expert annotations), Silver (layperson annotations), and Bronze (automatic annotations). The training split includes all folds, while the development and test sets contain only expert-curated data. The GutBrainIE collection, along with the annotation schema and guidelines, is publicly available on Zenodo[1]. Table 2 summarizes the distribution of documents and annotations across the folds.

Table 2
GutBrainIE dataset statistics.

Collection	No. Docs	No. Ents	Ents/Doc	No. Rels	Rels/Doc
Silver	499	15,275	30.61	10,616	21.27
Gold	208	5,192	24.96	1,994	9.59
Platinum	111	3,638	32.77	1,455	13.11
Dev	40	1,117	27.93	623	15.58
Test	40	1,237	30.92	777	19.42
Manual	898	26,459	29.46	15,465	17.22
Automatic (Bronze)	749	21,420	28.51	8,533	10.90
Overall	1,647	47,879	29.03	23,998	14.35

2.1. Comparison with Existing IE Collections

To better contextualize the features and strengths of the GutBrainIE collection, we compare it against several widely used general-purpose and biomedical IE datasets, as shown in Table 3. In this analysis, we excluded distantly supervised datasets to ensure a fair comparison in terms of reliability [28, 29].

GutBrainIE contains 1,647 documents, of which 898 are manually annotated, placing it among the largest manually curated biomedical IE collections, with a size comparable to NCBI Disease (793) and BioRED (600) [30, 21]. In terms of annotation density, GutBrainIE features an average of 29 entity mentions and 15 relations annotated per document. As shown in Table 3, these values are comparable to, or higher than, those reported for the other IE datasets included in our analysis.

Beyond size and density, a distinctive characteristic of GutBrainIE is the fine granularity of its annotation schema. Indeed, our corpus features 13 entity types and 17 relation predicates, almost doubling the level of granularity of BioRED (7 entity types, 8 relation predicates), which is the most fine-grained manually curated biomedical IE dataset presented to date in the literature [21]. Another distinctive feature of GutBrainIE is its tiered annotation quality structure, which organizes annotations

[1] https://zenodo.org/records/16845409

according to their reliability. This design allows models to be trained or tuned by weighting or filtering annotations differently based on their quality, supporting training strategies such as reliability-aware loss weighting and selective denoising [31, 32, 33, 34, 35].

Table 3
Comparison of GutBrainIE with several widely used IE datasets. Annotation counts include both entities and relations for datasets supporting RE. The "*Task(s)*" column reports the tasks supported by each dataset. Red rows indicate to biomedical IE datasets, while those in green denote general-purpose IE datasets.

Dataset	Year	Task(s)	No. Docs	No. Entity Mentions	No. Entity Types	No. Relations	No. Relation Predicates.	No. Annotations
GutBrainIE (Man)	2025	NER. NEL. RE	898	26,459 (29.46 per doc)	13	15,465 (17.22 per doc)	17	41,924 (46.69 per doc)
GutBrainIE (Man+Auto)	2025	NER. NEL. RE	1,647	47,879 (29.07 per doc)	13	23,998 (14.57 per doc)	17	71,877 (43.64 per doc)
JNLPBA [36]	2004	NER	2404	59,963 (24.94 per doc)	5	–	–	59,963 (24.94 per doc)
EU-ADR [37]	2012	NER. RE	300	7,011 (23.37 per doc)	3	2,436 (8.12 per doc)	3	9,447 (31.49 per doc)
BioNLP-CG [38]	2013	NER. RE	600	21,683 (36.14 per doc)	4	917 (1.53 per doc)	4	22,600 (37.67 per doc)
NCBI Disease [30]	2014	NER. NEL	793	6,892 (8.69 per doc)	1	–	–	6,892 (8.69 per doc)
CHEMDNER [39]	2015	NER	10,000	84,355 (8.44 per doc)	1	–	–	84,355 (8.44 per doc)
BC5CDR [40]	2016	NER. NEL. RE	1500	12,850 (8.57 per doc)	2	3,116 (2.08 per doc)	2	15,966 (10.64 per doc)
NLM-Gene [22]	2021	NER. NEL	550	15,553 (28.28 per doc)	1	–	–	15,553 (28.28 per doc)
BioRED [21]	2022	NER. NEL. RE	600	20,419 (34.03 per doc)	6	6,503 (10.84 per doc)	8	26,922 (44.87 per doc)
DocRED (Man) [41]	2019	NER. NEL. RE	5,053	98,533 (19.50 per doc)	6	56,798 (11.24 per doc)	96	155,331 (30.74 per doc)
HacRED [42]	2021	NER. NEL. RE	9,231	98,772 (10.70 per doc)	10	56,798 (6.15 per doc)	26	155,570 (16.85 per doc)
Re-DocRED [43]	2022	NER. NEL. RE	4,053	78,628 (19.40 per doc)	6	120,664 (29.77 per doc)	96	199,292 (49.17 per doc)

2.2. Benchmark Tasks

By design, the GutBrainIE corpus supports four complementary IE tasks of increasing complexity: NER, NEL, Mention-level RE (M-RE), and Concept-level RE (C-RE). The NER task requires the identification and classification of entity mentions into one of the 13 entity types defined in the annotation schema. Each mention is represented by its location in the document (title or abstract), character offsets, text span, and entity label. A predicted mention is considered correct when all these fields exactly match a corresponding ground-truth annotation. The NEL task extends NER by additionally requiring the linkage of each entity mention to a concept identifier (Uniform Resource Identifier (URI)) drawn from one of the reference vocabularies adopted in the collection. In this task, a prediction is counted as correct only if it matches the ground truth in all NER fields and in the associated concept URI. The M-RE task focuses on the extraction of relations between entity mentions. Each relation is represented as a 5-tuple comprising subject text span, subject label, relation predicate, object text span, and object label. Predicted relations are considered correct only if they fully match a ground truth tuple. The C-RE task mirrors M-RE but evaluates relations at the concept level rather than at the surface-form level. Relations are therefore represented as tuples composed of subject URI, subject label, predicate, object URI, and object label. By abstracting from surface forms, this formulation normalizes lexical variation across mentions referring to the same concept, allowing relations expressed through different textual realizations to be evaluated as equivalent. As a result, C-RE focuses on the semantic correctness of extracted relations and better reflects the needs of downstream applications such as knowledge base construction. Correctness is assessed in the same way as for M-RE.

In terms of evaluation metrics, all tasks are evaluated under Precision (P), Recall (R), and F1-score (F_1), with both macro- and micro-averaging. The label set corresponds to the 13 entity types for NER and NEL, and to the 55 valid relation triples induced by the annotation schema for M-RE and C-RE. Across all tasks, micro-averaged F_1 is used as the main reference metric, as it provides a more balanced evaluation for the effectiveness of IE systems while accounting for class imbalance.

2.2.1. Internal and External Validation

To validate the utility and the intended progression of difficulty in the tasks supported by the GutBrainIE benchmark, we conducted internal and external experiments. Internal validation was performed by converting the pipeline used for producing automatic annotations into a baseline system that supports all four tasks. For NER, we employed GLiNER fine-tuned from the NuNERZero checkpoint on the manually annotated subsets of the corpus [18, 44]. During inference, we applied a confidence threshold of 0.6 and a post-processing strategy for merging adjacent or overlapping spans. Predicted entities

were linked to concepts through the same three-stage NEL procedure employed during annotation. For RE we used ATLOP, trained using manually annotated data as primary supervision and automatically annotated data as distant supervision. Predicted relations were then processed to retain only triples compatible with the annotation schema. Baseline results across all four tasks are reported in Table 4.[2]

The benchmark was further validated externally by proposing the NER and M-RE tasks in an international evaluation campaign [15]. Participants were provided with the GutBrainIE dataset, its annotation schema and guidelines, and the baseline system implementation. In developing their systems, participants were free to adopt any architectural choice, training strategy, or external resource. A total of 17 teams joined the shared task, with 15 participating in NER (101 runs submitted) and 12 in M-RE (95 runs submitted). Considering micro-F_1 as the reference metric, 8 teams outperformed the baseline on NER (38 runs) and 5 on M-RE (24 runs). These results highlight both the competitiveness of the baseline and the clear room for methodological improvement. The systems used to produce the runs submitted to this campaign also provided useful insights into the current state of biomedical IE. For NER, the best results were obtained with supervised fine-tuning of biomedical transformer models such as PubMedBERT and BioBERT [25, 45]. For M-RE, most approaches framed the task as a supervised classification problem over pairs of entity mentions that could potentially be in a relation under the GutBrainIE annotation schema. Similarly to what was noticed for NER, top-performing systems employed supervised fine-tuning of biomedical transformers. For both tasks, performance improvements were obtained through ensemble strategies, negative sampling, and class-weighted loss functions. In contrast, incorporating the Bronze fold during training yielded limited and inconsistent gains. This suggests that, in fine-grained IE settings where label precision is crucial, the noise introduced by automatically generated annotations tends to reduce the potential benefits of additional training data. A few teams explored approaches based entirely on Large Language Models (LLMs), but these methods performed substantially worse than supervised systems.

Taken together, the internal and external results confirm that GutBrainIE provides a robust and challenging IE benchmark. The results validated the intended progression in complexity across tasks and underlined the importance of high-quality curation for obtaining reliable biomedical IE systems.

Table 4
Baseline results across the four tasks supported by the GutBrainIE benchmark.

	Macro-avg			Micro-avg		
Task	P	R	F_1	P	R	F_1
NER	0.69	0.71	0.70	0.76	0.82	0.79
NEL	0.40	0.41	0.39	0.50	0.54	0.52
M-RE	0.35	0.18	0.21	0.50	0.25	0.33
C-RE	0.30	0.18	0.20	0.38	0.20	0.27

3. The GutBrainIE Knowledge Base: GutBrainKB

To fully leverage the knowledge embedded in the GutBrainIE collection, it is necessary to extract, interpret, and integrate its content into a structured representation. A natural solution for this purpose is the use of KBs, which model entities as nodes and relations as edges in a machine-readable graph that supports efficient exploration and querying. We defined the KB derived from the GutBrainIE collection as GutBrainKB [17]. To support the representation of document- and sentence-level features of the annotations, as well as document metadata, we introduced the HEREDITARY Gut-Brain Ontology, which models the structural elements of the corpus.[3]

For the KB population process, we used the subset of expert-curated annotations from the GutBrainIE collection, as these already include concept-level links, essential for normalizing entity mentions and

[2]The full baseline implementation, along with the official evaluation scripts, is available at https://github.com/MMartinelli-hub/GutBrainIE_2026_Baseline.
[3]https://hereditary.dei.unipd.it/ontology/gutbrain/

constructing the KB. Starting from these, we extracted RDF triples capturing both the semantic relationships among entities identified through RE annotations and the textual context in which each entity appears (e.g., sentence- and document-level features), resulting in a final KB of 79K triples [46]. To ensure consistency with the underlying ontology, the resulting RDF graph has been validated with SHACL (Shapes Constraint Language). In our case, SHACL shapes were used to verify that entities, relations, and contextual elements comply with the annotation and ontological schemas [47, 48].

Importantly, the KB construction pipeline is fully reusable and scalable: new documents from future GutBrainIE expansions can be processed using the same workflow to generate additional RDF triples, which can be seamlessly integrated into the KB. As a result, the KB can grow over time while preserving schema consistency.

3.1. GutBrainKB Web Interface

To facilitate the exploration of the knowledge captured in GutBrainKB, we developed a web application that allows users to easily access and navigate the KB without requiring familiarity or expertise with semantic query languages (e.g., SPARQL). This platform provides an intuitive interface for searching biomedical concepts related to the gut-brain axis (cf. Figure 3) and for inspecting the relationships supported by the scientific literature from which they were extracted (cf. Figure 4) [17].

To search and retrieve a specific KB entry, users can simply search for it through an autocomplete search bar. For each concept, the interface presents its ontology reference, name, definition, and associated vocabulary, together with the relations in which the concept participates. The system also shows the textual evidence supporting each extracted relation, allowing users to inspect the sentences and documents from which the information has been derived. Bibliographic metadata associated with the evidence, such as article title, authors, journal, publication year, and PubMed identifier, are also available, enabling users to trace back each extracted fact to its corresponding scientific publication.

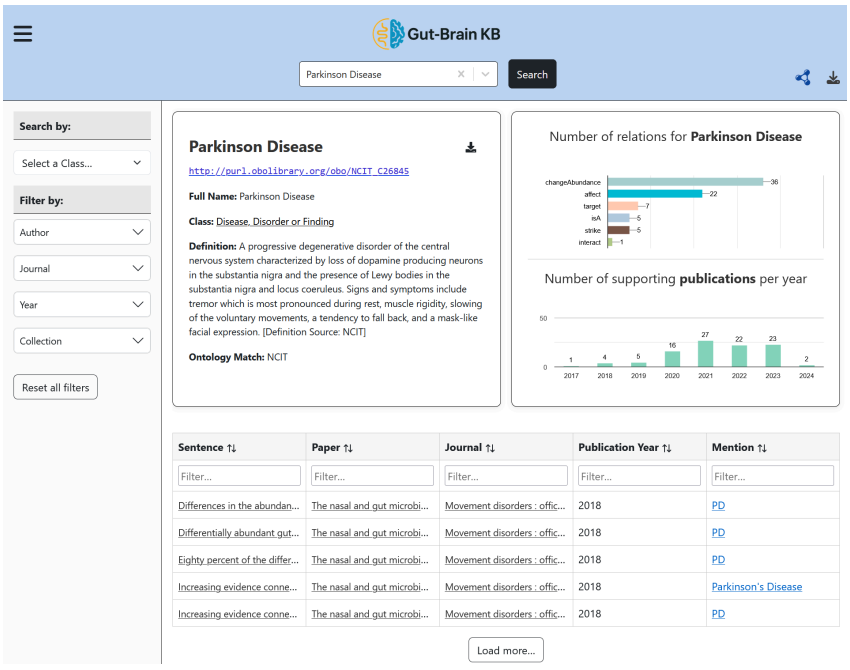


Figure 3: GutBrainKB User Interface showing the `Parkinson Disease` concept.

To support exploratory analysis, the interface includes filtering capabilities that allow users to narrow results by concept type, ontology class, publication source, author, or year. Relationships involving the selected concept are visualized both in a tabular and a graph-based format, highlighting the connections between the selected concept and other entities in the knowledge base. These visualizations help users quickly identify relevant associations and explore the structure of the extracted knowledge, enabling

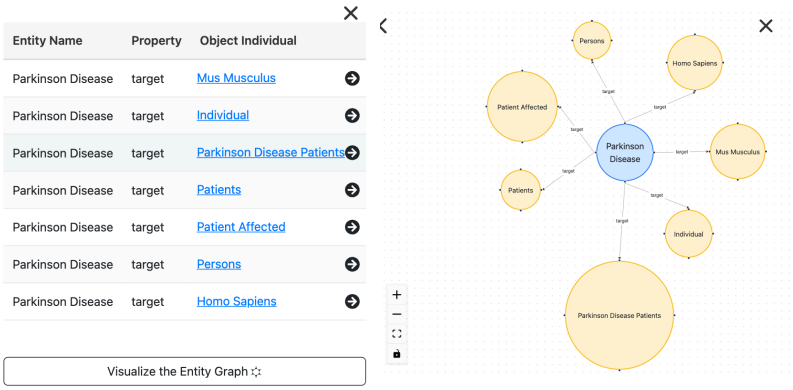


Figure 4: The panel on the left shows the list of relationships where *Parkinson disease* is the subject and *target* is the predicate. The right panel visualizes the corresponding Entity Graph.

step-by-step exploration of the KB by navigating into individual relationships and related entities. The platform also allows users to download summaries of the retrieved knowledge, including relationships and supporting textual evidence, in structured formats such as JSON or TTL, enabling integration with external tools.[4]

4. Discussion and Conclusions

In this paper, we presented GutBrainIE and GutBrainKB, two complementary resources designed to support research on biomedical IE and knowledge representation in the emerging domain of the gut–brain axis. GutBrainIE provides a large-scale corpus of PubMed abstracts annotated with entity mentions, concept-level links, and relations describing interactions between microbiota, biomedical entities, and neurological or psychiatric conditions. The collection supports 4 benchmark tasks of increasing complexity–NER, NEL, M-RE, and C-RE–for which we defined standardized evaluation procedures and implemented a baseline system to provide reference performance and facilitate reproducibility. The benchmark has been validated both internally and through external evaluation campaigns, confirming the relevance and difficulty of the proposed tasks. Several features of GutBrainIE make it a challenging and versatile benchmark for IE research. The dataset combines a relatively large scale with a high density of fine-grained annotations, providing rich supervision signals for model training and evaluation while reflecting the complexity of biomedical terminology in the gut-brain domain. Moreover, many relations span multiple sentences, requiring systems to capture long-range dependencies beyond sentence-level context. These properties make the benchmark suitable for broad research on IE methods capable of handling complex terminology, dense annotations, and cross-sentence relations.

Building on this annotated corpus, GutBrainKB demonstrates how curated IE resources can be leveraged to construct KBs that integrate biomedical knowledge extracted from scientific literature. The resulting KB provides a structured representation of the information and knowledge contained in the corpus, enabling advanced exploration and querying through the accompanying web interface.

Taken together, GutBrainIE and GutBrainKB showcase how the combination of curated datasets, evaluation benchmarks, and knowledge graph technologies can transform unstructured scientific literature into structured and interoperable knowledge resources.

Future work will focus on expanding the GutBrainIE corpus with recently published documents. The annotations produced for these new documents will then be incorporated into GutBrainKB through the same population pipeline, progressively enriching the knowledge graph while preserving consistency with the underlying ontology.

[4]The GutBrainKB web application is publicly available online at: https://hereditary.dei.unipd.it/app/gutbrainkb/

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Declaration on Generative AI

During the preparation of this work, the author used GPT-5.2 and Grammarly in order to: Grammar and spelling check. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the publication's content.

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A. Concept-Level Links in the GutBrainIE Collection

To support semantic normalization and concept-level RE, entities annotated in the Platinum and Gold collections have been linked to concepts from standardized biomedical vocabularies. We employed a total of six biomedical vocabularies and a custom-defined ontology. Each entity type is linked to a number of vocabularies ranging between 3 and 6. Table 5 reports the vocabularies used for each entity type and, for each pair of vocabulary and entity type, the number of distinct URIs employed.

Table 5
Biomedical vocabularies used for NEL. For each entity label, the Table lists the reference vocabularies, ordered accordingly to the priority considered in the NEL pipeline, and reports the number of unique URIs assigned for each vocabulary. *GBIE* denotes our custom-defined ontology.

Entity Label	Linked Vocabularies
Anatomical Location	UMLS (17), NCIT (70), GBIE (3)
Animal	UMLS (8), NCIT (7), NCBITaxon (5), GBIE (3)
Bacteria	UMLS (23), NCIT (6), NCBITaxon (136), MESH (34), OMIT (1), GBIE (6)
Biomedical Technique	UMLS (65), NCIT (16), OMIT (2), NCBITaxon (4), GBIE (53)
Chemical	UMLS (75), NCIT (94), CHEBI (209), OMIT (2), GBIE (14)
Dietary Supplement	NCIT (34), UMLS (11), CHEBI (13), NCBITaxon (4), OMIT (2), MESH (2), GBIE (3)
DDF	UMLS (179), NCIT (259), OMIT (36), NCBITaxon (1), GBIE (27)
Drug	UMLS (22), NCIT (9), CHEBI (34), OMIT (1), NCBITaxon (1), GBIE (3)
Food	UMLS (23), NCIT (17), GBIE (3)
Gene	UMLS (52), NCIT (42), OMIT (4), CHEBI (1), GBIE (13)
Human	UMLS (38), MESH (8), GBIE (24)
Microbiome	UMLS (2), NCIT (1), NCBITaxon (3), GBIE (15)
Statistical Technique	UMLS (35), NCIT (21), GBIE (23)

B. Relations Supported by the GutBrainIE Annotation Schema

GutBrainIE features annotations for 17 distinct relation predicates. Many predicates can connect multiple combinations of head and tail entity types, resulting in a total of 55 possible (*head, predicate, tail*) triples. Table 6 reports all relation predicates used in the GutBrainIE collection, represented as (*head, predicate, tail*) combinations according to the conceptual schema depicted in Figure 5.

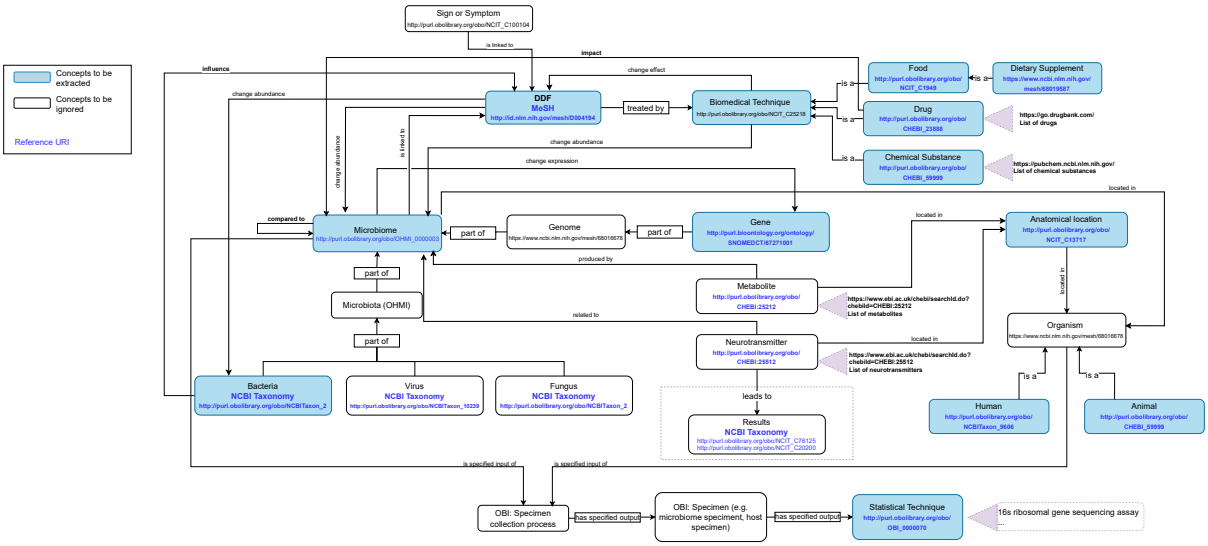


Figure 5: Conceptual schema illustrating the entity types and relation predicates defined in GutBrainIE. Blue rectangles denote annotated entity types, while white rectangles represent concepts considered during schema design but excluded from annotation due to their low frequency. Arrows indicate the valid relation predicates and their directions between entity types.

Table 6
Complete listing of the relations annotated in the GutBrainIE corpus, expressed as head-predicate-tail triples.

Head Entity	Tail Entity	Predicate
Anatomical Location	Human Animal	(1) Located in
Bacteria	Bacteria Chemical Drug	(2) Interact
Bacteria	DDF	(3) Influence
Bacteria	Gene	(4) Change expression
Bacteria	Human Animal	(1) Located in
Bacteria	Microbiome	(5) Part of
Chemical	Anatomical Location Human Animal	(1) Located in
Chemical	Chemical	(2) Interact (5) Part of
Chemical	Microbiome	(6) Impact (7) Produced by
Chemical Dietary Supplement Drug Food	Bacteria Microbiome	(6) Impact
Chemical Dietary Supplement Food	DDF	(3) Influence
Chemical Dietary Supplement Drug Food	Gene	(4) Change expression
Chemical Dietary Supplement Drug Food	Human Animal	(8) Administered
DDF	Anatomical Location	(9) Strike
DDF	Bacteria Microbiome	(10) Change abundance
DDF	Chemical	(2) Interact
DDF	DDF	(11) Affect (12) Is a
DDF	Human Animal	(13) Target
Drug	Chemical Drug	(2) Interact
Drug	DDF	(14) Change effect
Human Animal Microbiome	Biomedical Technique	(15) Used by
Microbiome	Anatomical Location Human Animal	(1) Located in
Microbiome	Gene	(4) Change expression
Microbiome	DDF	(16) Is linked to
Microbiome	Microbiome	(17) Compared to